Sequence

A function: $\mathbb{N}\to\mathbb{R}$, $n\mapsto f(n)=x_n$.

Denoted: $\{x_n\}_{n\in\mathbb{N}}$, $\{x_n\}_{n=1}^{\infty}$, $\{x_n\}$

Def 2.1: **Convergence:**

A sequence $\{x_n\}$ converges to $a\in\mathbb{R}$ if for any $\varepsilon>0$ there exists $N\in\mathbb{N}$ such that

$n\geq N \Rightarrow |x_n-a|<\varepsilon$

"$\forall\varepsilon>0,\ \exists N\in\mathbb{N}\ \ n\geq N\Rightarrow|a_n-a|<\varepsilon$"

We write: $x_n\to a$ or $\lim_{n\to\infty}x_n=a$.

$x_n\to a \Leftrightarrow x_n-a\to 0 \Leftrightarrow |x_n-a|\to 0$.

e.x. proving $1/n\to 0$.

$\forall\varepsilon>0\ \exists N=\frac{1}{\varepsilon}$.

So $\forall n\geq N\ |\frac{1}{n}-0|=\frac{1}{n}\leq\frac{1}{N}<\varepsilon$, proved.

e.x. Suppose $x_n\to 2$. prove that $\frac{2x_n+1}{x_n}\to\frac{5}{2}$.

$\varepsilon>0$. need N. such that $\forall\varepsilon>0$ $n\geq N \Rightarrow |\frac{2x_n+1}{x_n}-\frac{5}{2}|<\varepsilon$

$|\frac{2x_n+1}{x_n}-\frac{5}{2}|=|2+\frac{1}{x_n}-\frac{5}{2}|=|\frac{1}{x_n}-\frac{1}{2}|=|\frac{2-x_n}{2x_n}|<\varepsilon$ idea: find this

(1) Since $x_n\to 2$. for $\varepsilon=1$ $\exists N_1>0$ such that $n\geq N_1 \Rightarrow |x_n-2|<1 \Rightarrow x_n\in(1,3)$ $\Rightarrow x_n>1$

(2) Since $x_n\to 2$. for $\forall 2\varepsilon>0$ $\exists N_2>0$. $n\geq N_2 \Rightarrow |x_n-2|\leq 2\varepsilon$

So if $n\geq N=\max\{N_1,N_2\}$ then $x_n>1$ and $|x_n-2|<2\varepsilon$.

Such that $|\frac{2x_n+1}{x_n}-\frac{5}{2}|=\frac{|2-x_n|}{2|x_n|}<\frac{|x_n-2|}{2}<\frac{2\varepsilon}{2}=\varepsilon$.

A convergent sequence has only 1 limit

if $x_n\to a$ and $x_n\to b$ then $a=b$.

if $a\neq b$. $|a-b|=\varepsilon\neq 0$.

proof: if $a\neq b$. Assume $a>b$.

$|a-b|<|a-x_n|+|x_n-b|<\varepsilon=|a-b|$

$\varepsilon=a-b>0$.

$\exists N_1$ $n\geq N_1 \Rightarrow |x_n-a|<\frac{\varepsilon}{2}$.

$\exists N_2$ $n\geq N_2 \Rightarrow |x_n-b|<\frac{\varepsilon}{2}$.

if $n=\max\{N_1,N_2\}$. both above hold.

triangle inequality: $\varepsilon=|a-b|=|x_n-a+(b-x_n)|\leq|x_n-a|+|x_n-b|<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon$

$\Rightarrow \varepsilon<\varepsilon$. contradiction

Negating the definition

$x_n\to a$: $\forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall n\geq N: |x_n-a|<\varepsilon$.

$x_n\not\to a$: $\exists\varepsilon_0>0\ \forall N\in\mathbb{N}\ \exists n\geq N: |x_n-a|\geq\varepsilon_0$

$x_n\not\to a$: $\exists\varepsilon_0>0\ \forall N\in\mathbb{N}.\ n>N\Rightarrow|x_n-a|>\varepsilon_0$

e.x. prove $(-1)^n$ does not converge

proof: suppose $(-1)^n$ converges to a for some a.

for $\varepsilon=1$ $\exists N>0$: $\forall n\geq N$ $|(-1)^n-a|<1$.

But take $n=2N$ which is even.

$|(-1)^n-a|=|1-a|<1 \Rightarrow 0<a<2$

take $n=2N+1$ which is odd.

$|(-1)^n-a|=|-1-a|=|a+1|<1 \Rightarrow -2<a<0$

Since these two interval do not intersect: contradiction.

For subsequence remember: $n_k\geq k$

if $x_n\to a$. then every subsequence $\{x_{n_k}\}$ also converges to a

proof: $\forall\varepsilon>0\ \exists N>0$ $n\geq N \Rightarrow |x_n-a|<\varepsilon$.

Consider subsequence $\{x_{n_k}\}$

Since $n_k\geq k$ so if $k\geq N$ then $n_k\geq k\geq N$.

$k\geq N \Rightarrow n_k\geq N \Rightarrow |x_{n_k}-a|<\varepsilon$

$n>N\Rightarrow |x_n-a|<\varepsilon$

let $k:N$, $k>k\Rightarrow n_k>k>N\Rightarrow|x_{n_k}-a|<\varepsilon$

if every subsequences converge to a then the original sequence converge to a.

Bounded Sequence

M: upper bound.

N: lower bound

Thm 2.8 Every convergent sequence is bounded

Suppose $x_n\to a$ as $n\to\infty$

Take $\varepsilon=1$. $\exists N>0$. $n\geq N\Rightarrow|x_n-a|<1$.

$|x_n|=|x_n-a+a|\leq|x_n-a|+|a|<1+|a|$.

$\Rightarrow |x_n|\leq\max\{1+|a|,|a_1|,|a_2|,\cdots|a_n|\}$.

$n>N\Rightarrow|x_n-a|<1$.

$x_n<1+|a|$ for all $n>N$.

let $M=\max\{x_1,\cdots,x_{n-1},|a|+1\}$

$x_n\leq M$ for all $n\in\mathbb{N}$

Bounded Sequence may not converge. (e.g. $a_n=(-1)^n$)

Limit Theorem

Squeeze Theorem

(1) if $x_n\to a$ $y_n\to a$. and there is $N_0\in\mathbb{N}$ that $n\geq N_0 \Rightarrow x_n\leq w_n\leq y_n$ then $w_n\to a$.

(2) if $x_n\to 0$ and $\{y_n\}$ is bounded, then $x_ny_n\to 0$

$n>N_1\Rightarrow|x_n-a|<\varepsilon$.

$n>N_2\Rightarrow|y_n-a|<\varepsilon$

$N=\max\{N_0,N_1,N_2\}$

$n>N$

$a-\varepsilon<x_n\leq w_n\leq y_n<a+\varepsilon$

$|w_n-a|<\varepsilon \Rightarrow w_n\to a$

proof: Since $x_n\to a$ $y_n\to a$. $\forall\varepsilon>0\ \exists N_1,N_2>0$.

$n\geq N_1\Rightarrow|x_n-a|<\varepsilon\Rightarrow a-\varepsilon<x_n$ (1)

$n\geq N_2\Rightarrow|y_n-a|<\varepsilon\Rightarrow y_n<a+\varepsilon$ (2)

Thus: if $n\geq N_3: N_1+N_2+N_0$ then (1), (2) hold. and $x_n\leq w_n\leq y_n$

$a-\varepsilon<x_n\leq w_n\leq y_n<a+\varepsilon$. $\Rightarrow |w_n-a|<\varepsilon$

Hence w_n is converge to a.

proof: let $|y_n|<M$ $\forall n\in\mathbb{N}$ for some $M>0$.

Then $\forall\varepsilon>0$ Since $x_n\to 0$ $\exists N>0$

$n\geq N\Rightarrow|x_n|<\frac{\varepsilon}{M}$

So $n\geq N\Rightarrow|x_ny_n|\leq|x_n\cdot M|=M|x_n|<M\cdot\frac{\varepsilon}{|M|}=\varepsilon$.

$y_n<M$ $x_n<\frac{\varepsilon}{|M|}$

$|x_ny_n-0|<M\cdot\frac{\varepsilon}{|M|}=\varepsilon$

e.x. find: $\lim_{n\to\infty}2^{-n}\cos(n^3-n^2+n-1)$

(i) $2^{-n}\to 0$ as $n\to\infty$.

(ii) $\cos(n^3-n^2+n-1)$ is bounded $\Rightarrow \lim_{n\to\infty}2^{-n}\cos(n^3-n^2+n-1)=0$

$-1\leq\cos(n^3-n^2+n-1)\leq 1$

proof: $\forall\varepsilon>0$. $\exists N\in\mathbb{N}$, $N>\log\frac{1}{\varepsilon}$

$n\geq N\Rightarrow|a_n-a|=|\frac{1}{2^n}|=\frac{1}{2^n}<\frac{1}{2^N}<\frac{1}{2^{\log\frac{1}{\varepsilon}}}<\varepsilon$

$\therefore a_n\to 0$ by definition

Theorem 2.11: approximating a supremum or infimum

Let $\emptyset\neq E\subseteq\mathbb{R}$.

(i) If E has a finite supremum, there is a sequence $\{x_n\}\subseteq E$ such that $x_n\to\sup E$.

(ii) If E has a finite infimum, there is a sequence $\{y_n\}\subseteq E$ such that $y_n\to\inf E$.

E has suprenum let $m=\sup E$.

$\forall n\in\mathbb{N}$. $\exists x_n$. $m-\frac{1}{n}<x_n\leq m$

$\Rightarrow$ Squeze thm: $x_n\to\sup E$

proof: (i) Since E has a finite supremum

$\exists x_n\in E$. $s-\frac{1}{n}<x_n\leq s$ for any $n\in\mathbb{N}$

Since $s-\frac{1}{n}\to s$ as $n\to\infty$.

by squeeze theorem. $x_n\to s$ as $n\to\infty$.

(ii) $\exists x_n\in E$. $s\leq x_n<s+\frac{1}{n}$ $\Rightarrow$ $s+\frac{1}{n}\to s$ $\Rightarrow$ Squeeze theorem $\Rightarrow x_n\to s$

Theorem 2.12: algebra of finite limits

Suppose that $x_n\to a$ and $y_n\to b$, where $a,b\in\mathbb{R}$, and let $\alpha\in\mathbb{R}$. Then

(i) $x_n+y_n\to a+b$;

(ii) $\alpha x_n\to\alpha a$;

(iii) $x_ny_n\to ab$;

(iv) if $y_n\neq 0$ for every n and $b\neq 0$, then

$$\frac{x_n}{y_n}\longrightarrow\frac{a}{b}.$$

(i) $\forall\varepsilon>0$. $\exists N_1\in\mathbb{N}$. $n\geq N_1$ $|x_n-a|<\frac{\varepsilon}{2}$

$\exists N_2\in\mathbb{N}$ $n\geq N_2$ $|y_n-b|<\frac{\varepsilon}{2}$

$\Rightarrow$ let $N_3=\max\{N_1,N_2\}$

$n\geq N_3 \Rightarrow |x_n+y_n-(a+b)|=|x_n-a+(y_n-b)|\leq|x_n-a|+|y_n-b|<\varepsilon$.

$\Rightarrow x_n+y_n\to a+b$ as $n\to\infty$

(ii) $\forall\varepsilon>0$, $\exists N\in\mathbb{N}$. $n\geq N$, $|x_n-a|<\frac{\varepsilon}{|\alpha|}$

$\Rightarrow |\alpha x_n-\alpha a|=|\alpha||x_n-a|<|\alpha|\cdot\frac{\varepsilon}{|\alpha|}=\varepsilon$

$\Rightarrow \alpha x_n\to\alpha a$ as $n\to\infty$

(iii) since $x_n\to a$, $y_n\to b$. by theorem 2.8 x_n. y_n is bounded

$\Rightarrow |x_n|\leq M$. $|b|\leq M$ for some M

$\forall\varepsilon>0$. $\exists N_1\in\mathbb{N}$. $n\geq N_1$ $|x_n-a|<\frac{\varepsilon}{2M}$

$\exists N_2\in\mathbb{N}$ $n\geq N_2$ $|y_n-b|<\frac{\varepsilon}{2M}$

$|x_ny_n-ab|=|x_n(y_n-b)+b(x_n-a)|$

$\leq|x_n||y_n-b|+|b||x_n-a|$

$\leq M\cdot\frac{\varepsilon}{2M}+M\cdot\frac{\varepsilon}{2M}=\varepsilon$. $\Rightarrow x_ny_n\to ab$

(iv) ☆ first. we prove $\frac{1}{y_n}\to\frac{1}{b}$ as $n\to\infty$.

① $\exists N_1\in\mathbb{N}$. $n\geq N_1\Rightarrow|y_n-b|<\frac{b}{2}\Rightarrow y_n>\frac{b}{2}$

② $\exists N_2\in\mathbb{N}$ $n\geq N_2\Rightarrow|y_n-b|<\frac{b^2\cdot\varepsilon}{2}$

$n\geq\max\{N_1,N_2\}\Rightarrow|\frac{1}{y_n}-\frac{1}{b}|=\frac{|y_n-b|}{b|y_n|}<\frac{\frac{\varepsilon b^2}{2}}{b\cdot\frac{b}{2}}=\varepsilon$.

$\Rightarrow \frac{1}{y_n}\to\frac{1}{b}$ as $n\to\infty$.

$\Rightarrow$ By (iii), $\frac{x_n}{y_n}=x_n\cdot\frac{1}{y_n}\to\frac{a}{b}$

Prove: $\lim_{n\to\infty}\frac{n^3+n^2-1}{1-3n^3}=-\frac{1}{3}$ By thm 2.12.

$\lim_{n\to\infty}\frac{n^3+n^2-1}{1-3n^3}=\lim_{n\to\infty}\frac{1+\frac{1}{n}-\frac{1}{n^3}}{\frac{1}{n^3}-3}$

By definition of limit. we can prove $\frac{1}{n}\to 0$. $\frac{1}{n^3}\to 0$ as $n\to\infty$.

$\Rightarrow \lim_{n\to\infty}\frac{1+\frac{1}{n}-\frac{1}{n^3}}{\frac{1}{n^3}-3}=\lim_{n\to\infty}\frac{1+0-0}{0-3}=-\frac{1}{3}$.

Definition 2.14: infinite limits

Let $\{x_n\}$ be a real sequence.

(i) We write $x_n\to+\infty$ if, for every $M\in\mathbb{R}$, there is $N\in\mathbb{N}$ such that

$$n\geq N\implies x_n>M.$$

(ii) We write $x_n\to-\infty$ if, for every $M\in\mathbb{R}$, there is $N\in\mathbb{N}$ such that

$$n\geq N\implies x_n<M.$$

In either case the sequence diverges in $\mathbb{R}$, although it has a limit in the extended real line.

$x_n\to\infty$

$\forall M\in\mathbb{R}\ \exists N\in\mathbb{N}$.

$n>N\Rightarrow x_n>M$

Unbounded sets contain divergent sequences

Let $E\subseteq\mathbb{R}$.

(i) If $\sup E=+\infty$, there is a sequence $\{x_n\}\subseteq E$ such that $x_n\to+\infty$.

(ii) If $\inf E=-\infty$, there is a sequence $\{y_n\}\subseteq E$ such that $y_n\to-\infty$.

$\sup E=+\infty$

$\Rightarrow \forall n$, there are infinitely many $x_n>n$. Choose $x_n>n$ for every n.

then $\forall M\in\mathbb{R}$. $\exists N=M$.

$n>N\Rightarrow x_n>n>M$

$\Rightarrow x_n\to\infty$

The extended real number

Define: $\overline{\mathbb{R}}:=\mathbb{R}\cup\{-\infty,+\infty\}$

For $\emptyset\neq E\subseteq\mathbb{R}$: $\sup E=+\infty$ if E is unbounded above

$\inf E=-\infty$ if E is unbounded below

For $E=\emptyset$: $\sup E=-\infty$. $\inf E=\infty$.

proof: ① if $\sup E=\infty$, E is unbounded above

For every $n\in\mathbb{N}$, choose $x_n\in E$. with $x_n>n$ Construct a sequence

Given $M\in\mathbb{R}$ choose $N>M$. For $n\geq N$

$x_n>n\geq N>M$

By def 2.14. $x_n\to\infty$ as $n\to\infty$.

② if $\inf E=-\infty$. Choose $x_n<-n$

$x_n<-n\leq-N<-M \Rightarrow x_n\to-\infty$ as $n\to\infty$

Theorem 2.15: arithmetic with infinite limits

Suppose that $x_n\to+\infty$; the corresponding conclusions for $x_n\to-\infty$ are shown in parentheses.

(i) If $\{y_n\}$ is bounded below (respectively, bounded above), then

otherwise may $y_n\to-\infty$

$$x_n+y_n\to+\infty\qquad(\text{respectively, } x_n+y_n\to-\infty).$$

(ii) If $\alpha>0$, then

$$\alpha x_n\to+\infty\qquad(\text{respectively, } \alpha x_n\to-\infty).$$

(iii) If $y_n\geq M_0$ for some $M_0>0$ and every n, then

$$x_ny_n\to+\infty\qquad(\text{respectively, } x_ny_n\to-\infty).$$

(iv) If $\{y_n\}$ is bounded and $x_n\neq 0$ for every n, then

$$\frac{y_n}{x_n}\longrightarrow 0.$$

$\forall M\in\mathbb{R}$. $\exists N\in\mathbb{N}$.

$n>N\Rightarrow x_n>M-L$.

$y_n\geq L$ for all N.

$\Rightarrow x_n+y_n>M$

$n>N\Rightarrow x_n>\frac{M}{\alpha}$

proof: (i) suppose $y_n\geq L$ for all n Given $K\in\mathbb{R}$

$x_n\to+\infty\Rightarrow x_n>K-L$

$x_n+y_n>L+K-L=K \Rightarrow x_n+y_n\to+\infty$

(ii) $x_n\to+\infty\Rightarrow x_n>\frac{K}{\alpha}$ for any given $K\in\mathbb{R}$.

$\alpha x_n>\alpha\cdot\frac{K}{\alpha}=K \Rightarrow \alpha x_n\to+\infty$

(iii) $x_n\to\infty$. $\Rightarrow x_n>\max\{0,\frac{K}{M_0}\}$.

$x_ny_n>\frac{K}{M_0}\cdot M_0=K$. $\Rightarrow x_ny_n\to+\infty$

(iv) y_n is bounded $\Rightarrow |y_n|\leq B$.

$x_n\to+\infty$. $x_n>\frac{B}{\varepsilon}$ for any given $\varepsilon\in\mathbb{R}$

$|\frac{y_n}{x_n}-0|=\frac{|y_n|}{|x_n|}\leq\frac{B}{|x_n|}<\varepsilon$. $\Rightarrow \frac{y_n}{x_n}\to 0$.

The extended real line

Write

$$\overline{\mathbb{R}}:=\mathbb{R}\cup\{-\infty,+\infty\}.$$

The following conventions are consistent with Theorem 2.15:

Addition	Multiplication
$x+(+\infty)=+\infty$	$x(+\infty)=+\infty$ if $x>0$
$x+(-\infty)=-\infty$	$x(-\infty)=-\infty$ if $x>0$
$(+\infty)+(+\infty)=+\infty$	$x(+\infty)=-\infty$ if $x<0$
$(-\infty)+(-\infty)=-\infty$	$x(-\infty)=+\infty$ if $x<0$
	$(+\infty)(+\infty)=(-\infty)(-\infty)=+\infty$
	$(+\infty)(-\infty)=(-\infty)(+\infty)=-\infty$

Indeterminate forms

Expressions such as

$$(+\infty)-(+\infty),\qquad\frac{\infty}{\infty},\qquad 0\cdot\infty$$

are not assigned values. Different sequences producing the same symbolic form can have different limits, so each case requires further analysis.

Corollary 2.16: extended-real limit arithmetic

Let $x_n\to x$ and $y_n\to y$, where $x,y\in\overline{\mathbb{R}}$.

(i) $x_n+y_n\to x+y$, provided $x+y$ is not of the form $(+\infty)+(-\infty)$.

(ii) For $\alpha\in\mathbb{R}$, $\alpha x_n\to\alpha x$, provided the product is not of the form $0\cdot(\pm\infty)$.

(iii) $x_ny_n\to xy$, provided the product is not of the form $0\cdot(\pm\infty)$.

Theorem 2.17: comparison theorem

Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences. If there is $N_0\in\mathbb{N}$ such that

$$x_n\leq y_n\qquad(n\geq N_0),$$

then

$$\lim_{n\to\infty}x_n\leq\lim_{n\to\infty}y_n.$$

In particular, if $x_n\in[r,s]$ for every n and $x_n\to c$, then $c\in[r,s]$.

proof: Suppose $x_n\to a$. $y_n\to b$. and $a>b$ by contradiction

let $\delta=a-b>0$.

$\exists N_1.N_2\in\mathbb{N}$. $n\geq N_1\Rightarrow|x_n-a|<\frac{\delta}{2}$

$n\geq N_2\Rightarrow|y_n-b|<\frac{\delta}{2}$.

for $n\geq\max\{N_0.N_1.N_2\}$

$\delta=a-b=(a-x_n)+(x_n-y_n)+(y_n-b)$

$<\frac{\delta}{2}+0+\frac{\delta}{2}=\delta$.

Contradiction. hence $a\leq b$.

Strict inequalities may disappear in the limit

Even if $x_n<y_n$ for every sufficiently large n, one can conclude only that

$$\lim x_n\leq\lim y_n,$$

Can not take out equality

not that the inequality between the limits is strict.

For example,

$$\frac{1}{n}<\frac{1}{n}+\frac{1}{n^2}\qquad(n\in\mathbb{N}),$$

but

$$\lim_{n\to\infty}\frac{1}{n}=\lim_{n\to\infty}\left(\frac{1}{n}+\frac{1}{n^2}\right)=0.$$

Definition 2.18: monotone sequences

Let $\{x_n\}$ be a sequence of real numbers.

(i) It is *increasing* if

$$x_1\leq x_2\leq x_3\leq\cdots,$$

and *strictly increasing* if all these inequalities are strict.

(ii) It is *decreasing* if

$$x_1\geq x_2\geq x_3\geq\cdots,$$

and *strictly decreasing* if all these inequalities are strict.

(iii) It is *monotone* if it is either increasing or decreasing.

Some authors use *nondecreasing* and *nonincreasing* for the two nonstrict notions.

If $\{x_n\}$ is increasing and $x_n\to a$, we write

$$x_n\nearrow a.$$

Similarly, $x_n\searrow a$ means that $\{x_n\}$ is decreasing and converges to a. Notice that

$$x_n\nearrow a\iff -x_n\searrow -a.$$

Theorem 2.19: monotone convergence theorem

If $\{x_n\}$ is increasing and bounded above, or decreasing and bounded below, then $\{x_n\}$ converges to a finite limit.

Proof: $E:=\{x_n, n\in\mathbb{N}\}$. Since E is nonempty and bounded above.

By Completeness Theorem: $\exists\ a:=\sup E\in\mathbb{R}$

$\forall\varepsilon>0$, $\exists N\in\mathbb{N}$, $a-\varepsilon<x_N\leq a$

For every $n\geq N$. $a-\varepsilon<x_N\leq x_n\leq a$.

$a-\varepsilon<x_n\leq a \Rightarrow |x_n-a|<\varepsilon$. for every $n\geq N$

Hence. $x_n\to a$.

$x_n<x_{n+1}<\cdots$

$x_n<M$ for all n

$\forall\varepsilon>0$. $\exists N\in\mathbb{N}$.

$M-\varepsilon<x_N<M$

Since $x_n\nearrow M$

$n>N\Rightarrow M-\varepsilon<x_n<M$.

$|x_n-M|<\varepsilon\Rightarrow x_n\to M$

e.x. prove: if $|a|<1$. then $a^n\to 0$

proof: let $u_n:=|a|^n$ $u_{n+1}=|a|\cdot u_n$

thus. u_n is decreasing, and $u_n\geq 0\Rightarrow u_n$ is bounded below

By thm 2.19. $u_n\to L$ for some $L\geq 0$.

$u_{n+1}=|a|u_n \Rightarrow u_{n+1}\to L$.

$\lim_{n\to\infty}u_n=\lim_{n\to\infty}u_{n+1}\Rightarrow L=|a|L$.

Since $|a|\neq 0$. $L=0$

$|a^n|\to 0\Rightarrow a^n\to 0$

e.x. $a>0\Rightarrow a^{1/n}\to 1$.

(1) if $a=1$. $a^{1/n}=1$ for all n.

(2) if $a>1$. $u_n:=a^{1/n}$

Since $\frac{1}{n}>\frac{1}{n+1}$. u_n is decreasing.

Since $u_n>1$ for all n. u_n is bounded below.

$\Rightarrow u_n\to L$ for some $L\geq 1$

$u_{2n}\to L$ as well $\Rightarrow L^2=L\Rightarrow L=1$.

(3) if $0<a<1$. $a^{1/n}=\frac{1}{(1/a)^{1/n}}\to 1$ by (2)

Definition 2.22: nested sets

A sequence of sets $\{I_n\}_{n\in\mathbb{N}}$ is *nested* if

$$I_1\supseteq I_2\supseteq I_3\supseteq\cdots.$$

Theorem 2.23: nested interval property

Let $\{I_n\}$ be a nested sequence of nonempty, closed, bounded intervals. Then

$I_n\neq\emptyset$ for all n

$$E:=\bigcap_{n=1}^{\infty}I_n$$

is nonempty. If, in addition, the lengths satisfy $|I_n|\to 0$, then E consists of a single point.

Proof. Write

$$I_n=[a_n,b_n].$$

The nesting condition implies ① $a_n\leq a_{n+1}\leq b_{n+1}\leq b_n$.

$a_1\ a_2\cdots a_n\to\ b_1$

Thus $\{a_n\}$ is increasing and bounded above by b_1, while $\{b_n\}$ is decreasing and bounded below by a_1. By Theorem 2.19, there are $a,b\in\mathbb{R}$ such that

$$a_n\to a,\qquad b_n\to b.$$

Fix n. For every $m\geq n$,

$$a_n\leq a_m\leq b_m\leq b_n.$$

Letting $m\to\infty$ gives

$$a_n\leq a\leq b\leq b_n.$$

Consequently $[a,b]\subseteq I_n$ for every n, and hence

$$[a,b]\subseteq\bigcap_{n=1}^{\infty}I_n.$$

In particular, the intersection is nonempty.

If $|I_n|=b_n-a_n\to 0$, then

$$b-a=\lim_{n\to\infty}(b_n-a_n)=0,$$

so $a=b$. If $x\in E$, then $a_n\leq x\leq b_n$ for every n. Taking limits gives $a\leq x\leq b=a$, and therefore $x=a$. Thus $E=\{a\}$. □

$I_n=[a_n,b_n]$

$a_n\leq a_{n+1}\leq b_{n+1}\leq b_n$

$a_n\nearrow b_1$ $b_n\searrow a_1$ $\Rightarrow$ $a_n\to a$ for some a

$b_n\to b$ for some b

$\Rightarrow \bigcap_{n=1}^{\infty}I_n\supseteq[a,b]$, $[a,b]$ is nonempty

if $I_n=a_n-b_n\to 0$. then. $\forall x\in E$. $a_n\leq x\leq b_n$, Sqeeze thm $\Rightarrow x=a$

$a = b$

Bolzano-Weierstrass theorem

Every bounded sequence of real number has a convergent subsequence

proof: let $\{x_n\}$ be a bounded sequence, say $\{a_n\}\subseteq[a,b]$

cut by half

$a\quad a'\quad b$

Cut $[a,b]$ by half to get $[a,a']$, $[a',b]$

one of those two closed interval contains ∞ many x_n.

Say that interval is $I_1=[a_1,b_1]$

then $|I_1|=\frac{b-a}{2}$

$I_1\subseteq I_0=[a,b]$

cut I_1 by half

one of these two closed interval contains ∞ many x_n.

Say that interval I_2 $|I_2|=\frac{b-a}{4}$.

$I_2\subseteq I_1\subseteq I_0$

continue in this fashion we get a nested sequence of closed interval

$\cdots I_m\subseteq I_{m-1}\subseteq I_{m-2}\subseteq\cdots\subseteq I_1\subseteq I_0$

$I_m=[a_m,b_m]$ $|I_m|=\frac{1}{2^m}(b-a)\to 0$.

each I_m contains ∞ many x_n

$\Rightarrow \bigcap_{m=1}^{\infty}I_m=\{x\}$.

Now take one element $x_{n_k}\in I_k$ $k\in\mathbb{N}$

$0\leq|x_{n_k}-x|\leq|I_k|=\frac{b-a}{2^k}\to 0$.

The sequence theorem gives $x_{n_k}\to x$

Cauchy Sequence

$\{x_n\}$ said to be cauchy $\Leftrightarrow$ $\forall\varepsilon>0\ \exists N\in\mathbb{N}$, $m,n\geq N\Rightarrow|x_m-x_n|<\varepsilon$.

remark: if $\{x_n\}$ is convergent then it is cauchy

$m,n>N$: $|x_m-x_n|<|x_m-a|+|x_n-a|<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon$

$\Rightarrow$ Cauchy

Theorem: $\{x_n\}$ is cauchy $\Leftrightarrow$ $\{x_n\}$ convergent to some point $x\in\mathbb{R}$ "$\Leftarrow$": proved "$\Rightarrow$"

Suppose $\{x_n\}$ is cauchy

① x_n is bounded $\Rightarrow$ $\exists\{x_{n_k}\}$ is converge to a. ② B-W theorem

$\Rightarrow$ ③ x_n is converge to a since $\{x_n\}$ is cauchy

for $\varepsilon=1$ $\exists N>0$ $m.n\geq N\Rightarrow|x_m-x_n|<1$.

$\Rightarrow |x_m-x_N|<1$ for any $m\geq N$

$|x_m|=|x_m-x_N+x_N|\leq|x_m-x_N|+|x_N|<1+|x_N|$

So $|x_m|\leq\max\{|x_1|,|x_2|,\cdots,|x_{N-1}|,1+|x_N|\}$ for any m

$\Rightarrow \{x_n\}$ is bounded

② $\forall\varepsilon>0$ $\exists k_0$ $k\geq k_0\Rightarrow|x_{n_k}-a|<\frac{\varepsilon}{2}$.

Since x_n is cauchy, $\exists N>0$ $m,n\geq N\Rightarrow|x_m-x_n|<\frac{\varepsilon}{2}$

☆ let $k\geq k_0$ such that $n_k\geq N$

Then $\forall n\geq N$ we have

$|x_n-a|\leq|x_n-x_{n_k}|+|x_{n_k}-a|<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon$.

if Cauchy $n,m\geq N$. $|x_m-x_n|<1$.

fixed $n=N$. $\forall m>N$. $|x_m|<|x_N|+1$

let $M=\max\{x_1,x_2,\cdots,x_{N-1},|x_N|+1\}$

$|x_n|<M$ for all $n\Rightarrow x_n$ is bounded

then B-W thm: x_n has a convergent subseq $\{x_{n_k}\}$

$x_{n_k}\to a$: $\forall\varepsilon$. $\exists N_1$ $k>N_1\Rightarrow|x_{n_k}-a|<\frac{\varepsilon}{2}$.

Cauchy: $\exists N_2$. $m,n>N_2\Rightarrow|x_m-x_n|<\frac{\varepsilon}{2}$

let $N=\max\{N_1,N_2\}$

$n>N\Rightarrow|x_n-a|<|x_n-x_{n_k}|+|x_{n_k}-a|<\varepsilon$

$\{x_n\}$, $|x_n-x_{n+1}|\leq\frac{1}{2^n}\Rightarrow x_n$ is convergent

$|x_n-x_m|=|x_n-x_{n+1}+x_{n+1}-x_{n+2}\cdots+x_{m-1}-x_m|$

$\leq|x_n-x_{n+1}|+|x_{n+1}-x_{n+2}|+\cdots+|x_{m-1}-x_m|$

$\leq\frac{1}{2^n}+\frac{1}{2^{n+1}}+\cdots+\frac{1}{2^{m-1}}$

$\leq\frac{1}{2^n}$

remark: $\{x_n\}$ satisfying $x_{n+1}-x_n\to 0$ need not to be cauchy

$x_n=\log n$

$x_{n+1}-x_n=\log(n+1)-\log n=\log\frac{n+1}{n}\to\log 1=0$ as $n\to\infty$

However, since $\log n$ doesn't converge. it is not cauchy

Limit Superior & inferior

$\{x_n\}_{n=1}^{\infty}$

$S_n=\sup_{k\geq n}x_k=\sup\{x_k: k\geq n\}$

$t_n=\inf_{k\geq n}x_k=\inf\{x_k: k\geq n\}$

e.x. $\{\frac{1}{n}\}_{n=1}^{\infty}$ $S_n=\sup_{k\geq n}x_k=\sup_{k\geq n}\frac{1}{k}=\frac{1}{k}$.

$t_n=\inf_{k\geq n}x_k=\inf_{k\geq n}\frac{1}{k}=0$

remark: if $A\subseteq B$, then:

$\Rightarrow \sup A\leq\sup B$. $\inf A\geq\inf B$

Since $\{x_k: k\geq n+1\}\subseteq\{x_k: k\geq n\}$

$\Rightarrow S_{n+1}\leq S_n$. $t_{n+1}\geq t_n$

$S_n\geq S_{n+1}\geq S_{n+2}\cdots$

$t_n\leq t_{n+1}\leq t_{n+2}\cdots$

$t_n\quad t_{n+1}\quad t_{n+2}\ \cdots\ \leftarrow S_{n+2}\quad S_{n+1}\quad S_n$

$\Rightarrow \lim_{n\to\infty}S_n=a$. $(a\in\mathbb{R}\cup\{+\infty\})$

$\lim_{n\to\infty}t_n=b$ $(b\in\mathbb{R}\cup\{-\infty\})$

Limit supremum/infimum

$\limsup_{n\to\infty}x_n:=\lim_{n\to\infty}(\sup_{k\geq n}x_k)$

$\liminf_{n\to\infty}x_n:=\lim_{n\to\infty}(\inf_{k\geq n}x_k)$

e.x. for $x_n=(-1)^n$

$\limsup_{n\to\infty}x_n=\lim_{n\to\infty}(\sup_{k\geq n}(-1)^k)=1$. $=\sup\{-1,1\}=1$.

$\liminf_{n\to\infty}x_n=\lim_{n\to\infty}(\inf_{k\geq n}(-1)^k)=-1$ $\inf\{-1,1\}=-1$

e.x. $x_n=1+1/n$.

$\limsup_{n\to\infty}x_n=\lim_{n\to\infty}(\sup_{k\geq n}(1+1/n))=\lim_{n\to\infty}(1+\frac{1}{n})=1$.

$\liminf_{n\to\infty}x_n=\lim_{n\to\infty}(\inf_{k\geq n}(1+1/n))=\lim_{n\to\infty}1=1$.

Thm suppose $\{x_n\}$ is a bounded sequence

$\limsup_{n\to\infty}x_n=a\in\mathbb{R}$ $\liminf_{n\to\infty}x_n=b\in\mathbb{R}$

(1) if $r_1\leq x_n\leq r_2$ $\forall n\in\mathbb{N}$. then $b\leq r_1\leq r_2\leq a$

proof: $r_1\leq x_n\leq r_2$ $\forall n\in\mathbb{N}$.

$a=\limsup_{n\to\infty}x_n=\lim_{n\to\infty}(\sup_{k\geq n}x_k)\leq r_2$

$b=\liminf_{n\to\infty}x_n=\lim_{n\to\infty}(\inf_{k\geq n}x_k)\geq r_1$

(2) $\exists$ subsequence. $\{x_{n_k}\}$, $\{x_{n_k'}\}$. such that.

$\lim_{k\to\infty}x_{n_k}=a$ $\lim_{k\to\infty}x_{n_k'}=b$

proof: since $\limsup_{n\to\infty}x_n=a$.

$S_n:=\sup\{x_k: k\geq n\}$, $S_n\searrow a$.

$\{n_k'\}$.

bounded

$x_{n_1}\leq S_{n_0}$

For $n_0=0$ $S_1=S_{n_0+1}=\sup\{x_k: k\geq 1\}$.

$\exists n_1\in\mathbb{N}$ $S_{n_0+1}-1<x_{n_1}\leq S_{n_0+1}$.

$\exists n_2\geq n_1+1$ $S_{n_1+1}-\frac{1}{2}<x_{n_2}<S_{n_1+1}$

$\exists n_3\geq n_2+1$ $S_{n_2+1}-\frac{1}{3}<x_{n_3}<S_{n_2+1}$

$\cdots$

$S_{n_{k-1}+1}-\frac{1}{k}<x_{n_k}\leq S_{n_{k-1}+1}$, $\forall k\in\mathbb{N}$

$k\to\infty$. $\lim_{n\to\infty}S_n=a\Rightarrow S_{n_k+1}\to a$ as $k\to\infty$.

Sequeeze thm: $\lim_{k\to\infty}x_{n_k}=a$

(3) if $\{x_{n_k}\}$ is a convergent subsequence of $\{x_n\}$.

Then $b\leq\lim_{k\to\infty}x_{n_k}\leq a$

proof: Suppose $\lim_{k\to\infty}x_{n_k}=x$. N ⓝ$_k$

$k\to\infty$. $x_{n_k}\to x$

$\forall N\in\mathbb{N}$, $\exists k_N>0$. $k\geq k_N\Rightarrow n_k\geq N$.

$\Rightarrow \inf\{x_n: n\geq N\}\leq x_{n_k}\leq\sup\{x_n: n\geq N\}$ $\forall k$ $k\to\infty$.

$\Rightarrow \inf\{x_n: n\geq N\}\leq x\leq\sup\{x_n: n\geq N\}$ $N\to\infty$.

$b\leq x\leq a$

(4) The sequence $\{x_n\}$ converges $\Leftrightarrow a=b$.

1. $x_n\to x\Rightarrow x_{n_k}$. $x_{n_k'}\to x\Rightarrow a=b=x$

2. if $a=b$: $t_n\leq x_n\leq S_n\Rightarrow$ Sequeeze thm: $x_n\to a$.

Theorem: The following are equivalent:

(i) $\{x_n\}$ is not bounded above

(ii) $\limsup_{x\to\infty}x_n=\infty$

(iii) there is a subsequence $\{x_{n_k}\}$ such that $\lim_{k\to\infty}x_{n_k}=+\infty$.

(i) → (ii):

x_n is unbounded above $\Rightarrow$ $\{x_n: n\geq k\}$ is unbounded above

$\Rightarrow \sup\{x_n: n\geq k\}=+\infty$ for all $k\in\mathbb{N}$

(ii) → (iii):

Choose n_1 so that $x_{n_1}>1$.

Choose $n_j>n_{j-1}$ & $x_{n_{j+1}}>j+1$

$\Rightarrow x_{n_j}\to\infty$

(iii) → (i) if $\{x_n\}$ is bounded above, then every subsequence is bounded above. then no subseq could tend to ∞

if $x_n\leq y_n$ for all sufficient large n

then $\limsup_{n\to\infty}x_n\leq\limsup_{n\to\infty}y_n$.

$\liminf_{n\to\infty}x_n\leq\liminf_{n\to\infty}y_n$

for $k\geq N$. $x_k\geq y_k\Rightarrow$ $n\geq N\Rightarrow$ $\sup_{k\geq N}x_k\leq\sup_{k\geq N}y_k$ $\inf_{k\geq N}x_k\leq\inf_{k\geq N}y_k$

$n\to\infty$: $\limsup_{n\to\infty}x_n\leq\limsup_{n\to\infty}y_n$.

$\liminf_{n\to\infty}x_n\leq\liminf_{n\to\infty}y_n$

Nested Interval

$\bigcap_{n=1}^{\infty}I_n=[a,b]$ nonempty

$|I_n|\to 0\Rightarrow\bigcap_{n=1}^{\infty}=\{d\}$

BW Theorem

Bounded Seq has a Convergent subseq

Monotone $x_n\nearrow a$, $x_n\searrow a$

if $x_n\nearrow$ & x_n is bounded above $\Rightarrow$ Converge to Sm a

Comparison thm:

N_0: $n>N_0$. $x_n\leq y_n$

$\lim x_n\leq\lim y_n$

still $x_n<y_n$

Convergent Seq has only 1 limit

Convergence

$x_n\to a$ $\forall\varepsilon>0$. $\exists N\in\mathbb{N}$. $n>N\Rightarrow|x_n-a|<\varepsilon$

$x_n\to\infty$ $\forall M\in\mathbb{R}$. $\exists N\in\mathbb{N}$. $n>N\Rightarrow x_n>M$

$x_n\not\to a$. $\exists\varepsilon_0>0$ $\forall N\in\mathbb{N}$. $n>N$ $|x_n-a|\geq\varepsilon_0$

Seq Convergent $\Leftrightarrow$ all subseq converge

arithmetic

t_n

Comparison $n>N\Rightarrow x_n\leq y_n$

$\limsup x_n\leq\limsup y_n$

$\liminf x_n\leq\liminf y_n$

Squeze thm

1. $x_n\leq w_n\leq y_n$; $a\to a\leftarrow a$

2. $x_n\to 0$. y_n bounded $\Rightarrow x_ny_n\to 0$

inf / sup

if has inf / sup (can $=\infty/-\infty$)

has subseq → inf / sup

Converge ⇒ Bounded

Bounded ⇏ Converge ($\sin x$)

infinite

"$=$" $\limsup_{n\to\infty}x_n=+\infty$

x_n is unbounded

x_n has subseq $\to+\infty$

limit Superior

$S_n=\sup\{x_k: k\geq n\}$

$t_n=\inf\{x_k: k\geq n\}$

$\Rightarrow \limsup_{n\to\infty}x_n=\lim_{n\to\infty}S_n$. $\liminf_{n\to\infty}x_n=\lim_{n\to\infty}t_n$

$S_n\geq S_{n+1}$. $t_n\leq t_{n+1}$

Caichy Seq → Def. $m,n>N\Rightarrow|x_m-x_n|<\varepsilon$

Cauchy ⇔ Convergent

properties if f: bounded

$\limsup_{n\to\infty}x_n=b$. $\liminf_{n\to\infty}x_n=a$

① if $r_1\leq x_n\leq r_2$ $\forall n$ then $r_1\leq a\leq b\leq r_2$

② ∃ subseq x_{n_k}. $x_{n_k'}$: $x_{n_k}\to a$. $x_{n_k'}\to b$

③ ∀ Convergent subseq, $x_n\to m$. $a\leq m\leq b$

④ x_n converge ⇔ $a=b$

Exercise 1. If $\lim_{n\to\infty}a_n=a$, then

$$\lim_{n\to\infty}\frac{a_1+a_2+\cdots+a_n}{n}=a.$$

if $a=0$: Since $\lim_{n\to\infty}a_n=a$. $\forall\varepsilon>0\ \exists N\in\mathbb{N}$. $n>N\Rightarrow|a_n|<\frac{\varepsilon}{2}$.

let $S_N=\sum_{n=1}^{N}a_n$. $\exists n_0\in\mathbb{N}$. $n>n_0\Rightarrow\frac{S_N}{n}<\frac{\varepsilon}{2}$

for $N_1=\max\{n_0,N\}$: $n>N_1$.

$|\frac{a_1+a_2+\cdots+a_n}{n}|<|\frac{S_N}{n}|+|\frac{a_{N+1}+\cdots+a_n}{n}|<\frac{\varepsilon}{2}+\frac{(n-N+1)}{n}\cdot\frac{\varepsilon}{2}<\varepsilon$.

$\frac{a_1+a_2+\cdots+a_n}{n}\to a$ as $n\to\infty$.

for general case: $a\neq 0$.

let. $b_n=a_n-a$ $b_n\to 0$ as $n\to\infty$, as shown above: $\lim_{n\to\infty}\frac{b_1+b_2+\cdots+b_n}{n}=0$.

$\Rightarrow \frac{a_1-a+a_2-a+\cdots+a_n-a}{n}\to 0$. as $n\to\infty$

$\Rightarrow \frac{a_1+a_2+\cdots+a_n}{n}-a\to 0$ as $n\to\infty$.

$\Rightarrow \lim_{n\to\infty}\frac{a_1+a_2+\cdots+a_n}{n}=a$.

Exercise 2. Prove that $\mathbf{R}$ is uncountable.

Method 1: Cantor's theorem: $(0,1)$ is uncountable.. $(0,1)\subseteq\mathbb{R}\Rightarrow\mathbb{R}$ is uncountable

Method 2: Suppose $\mathbb{R}$ is countable

Then: $\mathbb{R}=\{x_1,x_2,x_3,\cdots\}$.

Choose a closed interval $[a_1,b_1]$ with $x_1\notin[a_1,b_1]$

divide into 3 parts: $[a_1,\frac{2a_1+b_1}{3}]$, $[\frac{2a_1+b_1}{3},\frac{a_1+2b_1}{3}]$ $[\frac{a_1+2b_1}{3},b_1]$

At most two of them contains x_2. $\Rightarrow$ at least one does not contain x_2, denote it $[a_2,b_2]$

divide $[a_2,b_2]$ into 3 parts.

At least one subsequence does not contain x_3

⋮

$[a_{n+1},b_{n+1}]\subseteq[a_n,b_n]$, $x_{n+1}\notin[a_n,b_n]$

$b_n-a_n=(b_1-a_1)(\frac{1}{3})^{n-1}\to 0$ as $n\to\infty$.

$\Rightarrow \bigcap_{n=1}^{\infty}[a_n,b_n]=\xi$

$\Rightarrow \xi$ is a real number, $\xi\notin\{x_1,\cdots\}\Rightarrow$ contradiction

Exercise 3. If a sequence $\{x_n\}$ converges to a, then every subsequence $\{x_{n_k}\}$ also converges to a, that is,

$$\lim_{n\to\infty}x_n=a\Rightarrow\lim_{k\to\infty}x_{n_k}=a.$$

$\lim_{n\to\infty}x_n=a\Rightarrow\forall\varepsilon>0\ \exists N\in\mathbb{N}$. $n>N\Rightarrow|x_n-a|<\varepsilon$

let $K=N$. $k>K\Rightarrow n_k\geq k>N\Rightarrow|x_{n_k}-a|<\varepsilon$

$\Rightarrow x_{n_k}\to a$ as $n\to\infty$

Corollary 1. If a sequence $\{x_n\}$ has two subsequences $\{x_{n_k^{(1)}}\}$ and $\{x_{n_k^{(2)}}\}$ that converge to different limits, then $\{x_n\}$ diverges.

Example 1. The sequence $\left\{\left|\sin\frac{n\pi}{4}\right|\right\}$ diverges.

for $n_k=4k$. $x_{n_k}=0$. $x_{n_k}\to 0$ as $n\to\infty$

for $n_k=8k+2$. $x_{n_k}=1$. $x_{n_k}\to 1$ as $n\to\infty$

$\Rightarrow$ Diverge

Exercise 4. If $\{x_n\}$ is an unbounded sequence, then there exists a subsequence $\{x_{n_k}\}$ such that:

$$\lim_{k\to\infty}x_{n_k}=\infty.$$

x_n is an unbounded sequence $\Rightarrow\exists n_1$, $x_{n_1}>1$.

$\exists n_2>n_1$. $x_{n_2}>2$. $\cdots$ $\exists n_{k+1}>n_k$. $x_{n_{k+1}}>k+1$

$\Rightarrow x_{n_k}\to\infty$ as $n_k\to\infty$